

Can We Predict Hotel Booking Cancellations Before They Happen?

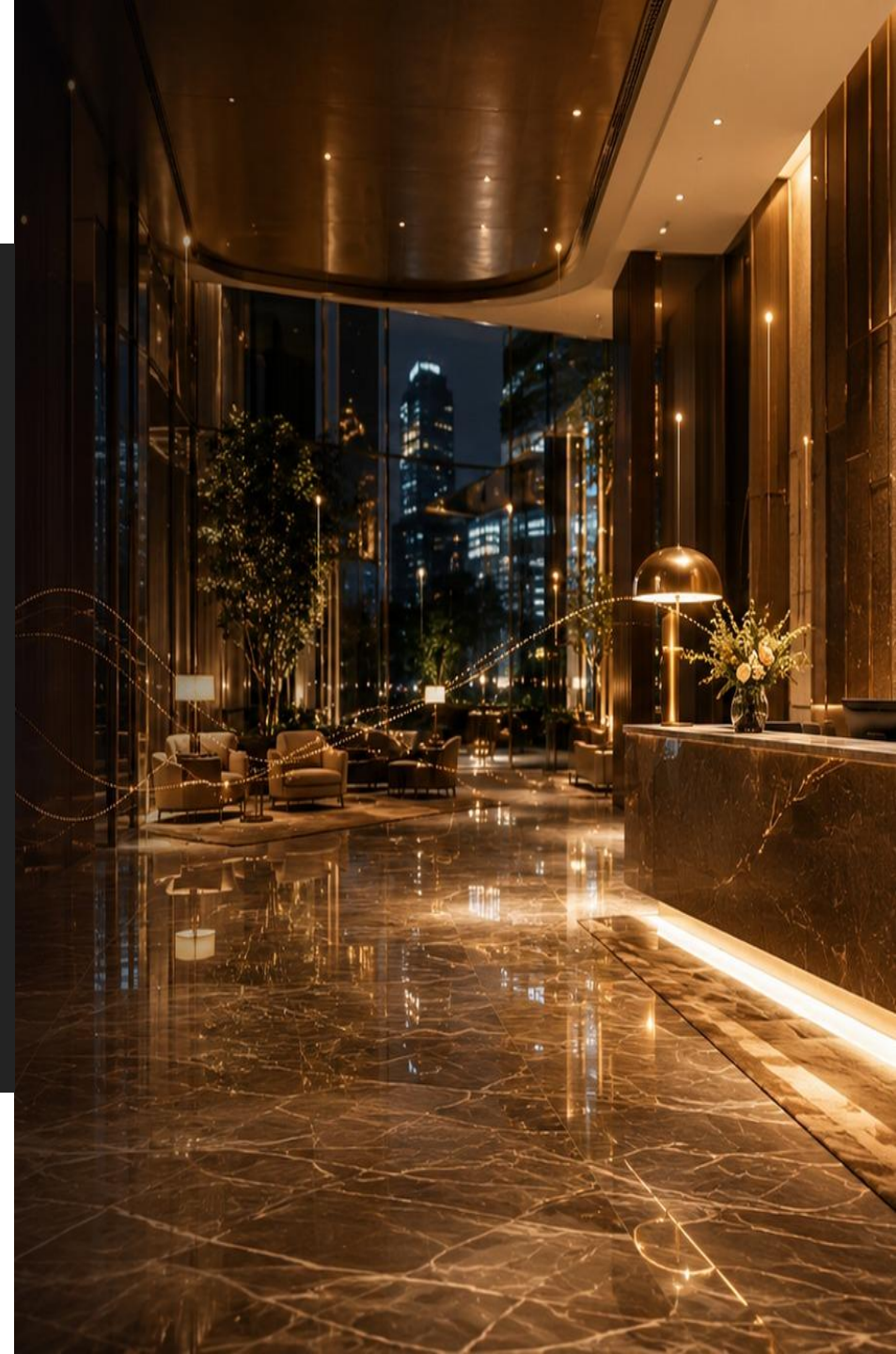
All charts + analysis from the Hotel Haven cancellation study

EDA • Risk drivers • Model storyline • Streamlit deployment

PREDICTIVE MODEL TO DETERMINE CUSTOMER BOOKING STATUS

PREDICTIVE MODEL TO DETERMINE CUSTOMER BOOKING STATUS

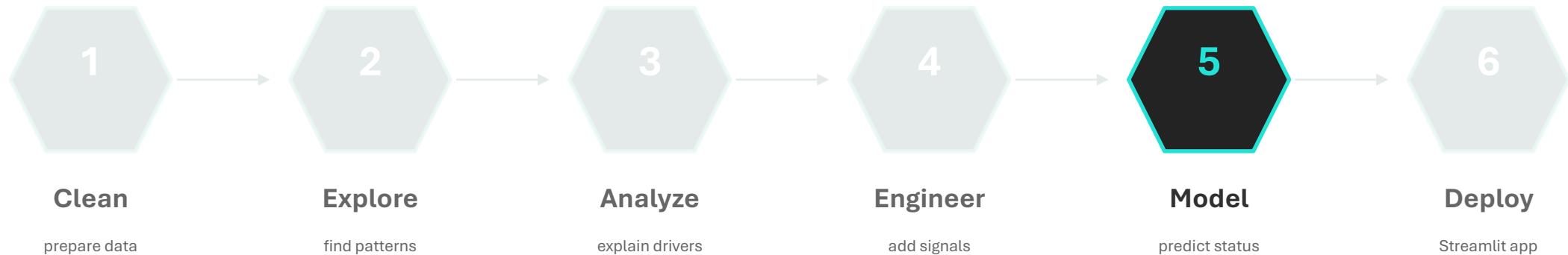
Uchechukwu Eze



PROJECT STORYLINE

The model follows a complete data science flow: from business problem to deployed prediction app.

Hotel Haven is a luxury hotel chain facing high cancellation rates. The aim is to understand booking behavior and predict cancellation risk before it damages revenue and planning.



The story is not just “we built a model.” The story is: “Wee found where cancellation risk starts, and gave the hotel a way to act earlier.”

High cancellations are not just lost bookings. They are lost control.

Hotel Haven needs to predict which customers are likely to cancel so it can protect revenue, allocate rooms better, and design targeted retention actions.

Clean

prepare the data for analysis

Analyze

identify booking patterns

Engineer

add useful model features

Predict

classify cancel vs not cancel

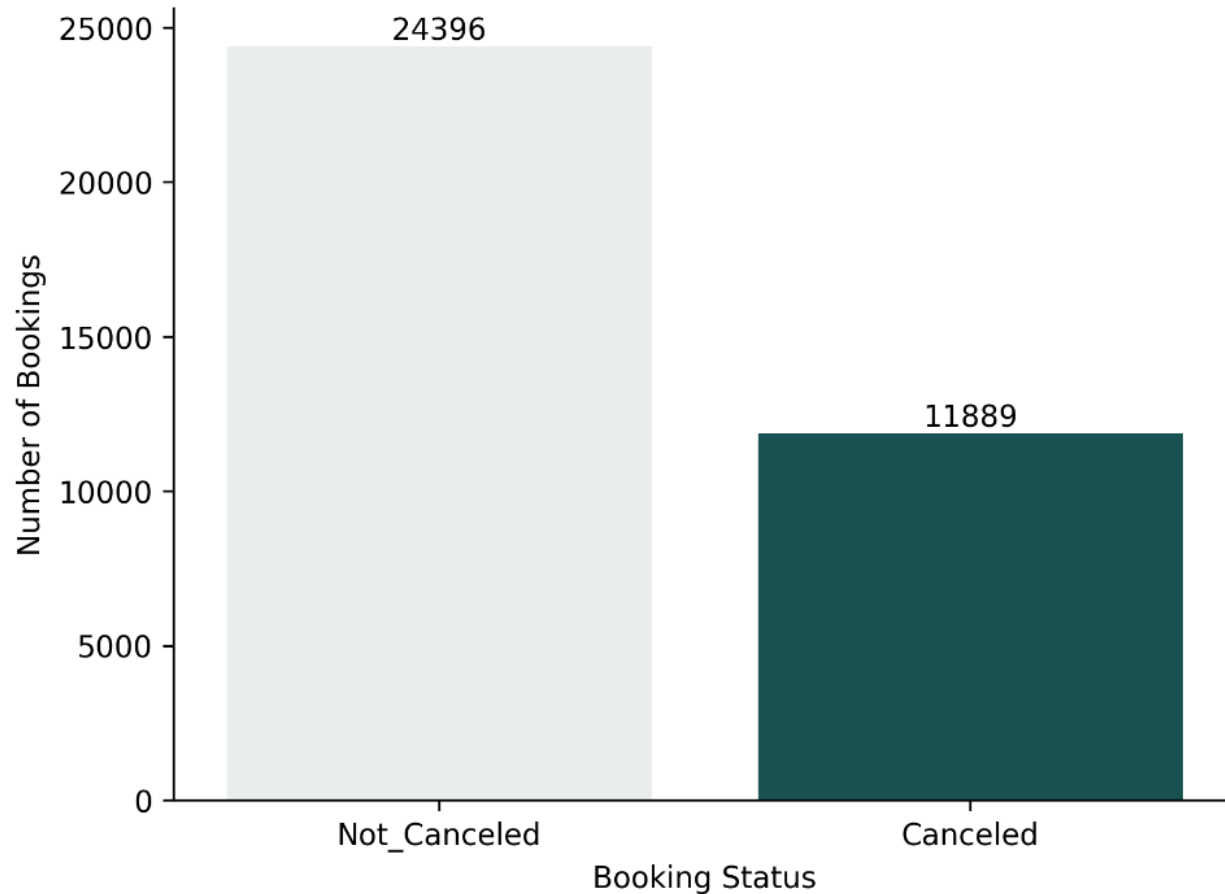
Deploy

enable user interaction



BOOKING STATUS DISTRIBUTION

The first look shows the cancellation problem is not small.



Booking status statistics is showing that over 67% (24396) of Hotel Haven have not yet canceled their booking tickets. With over 33% (11889) cancellation status

ANALYSIS

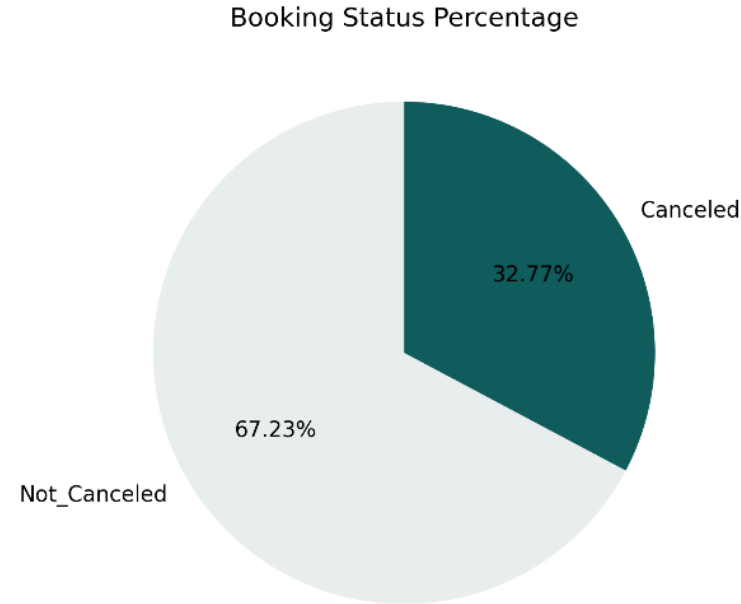
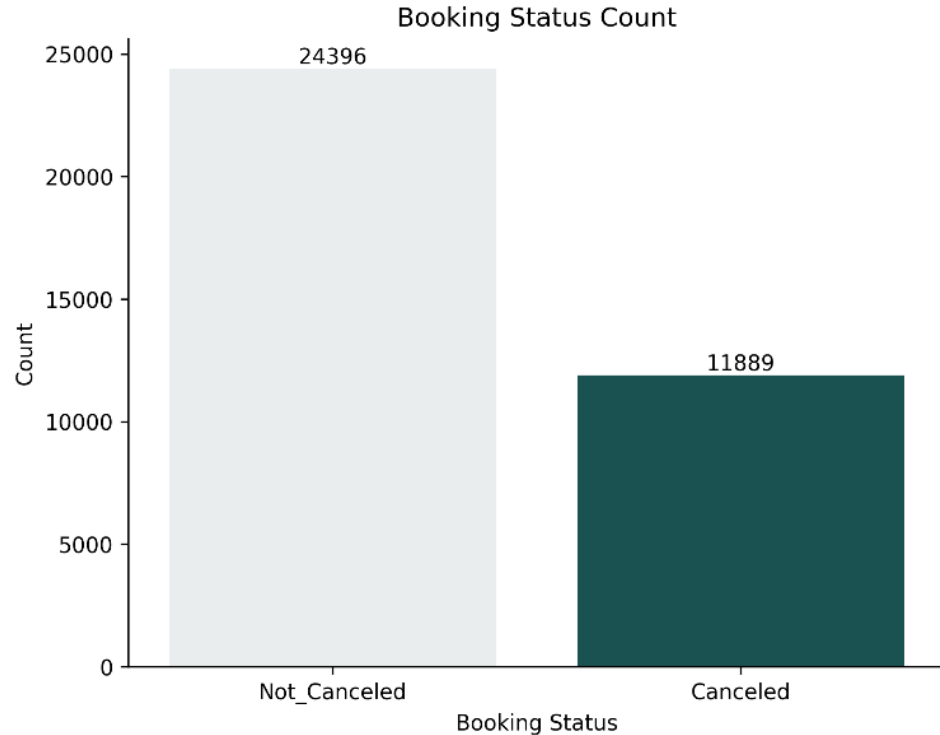
One in three bookings is at risk.

- 24,396 bookings were not canceled.
- 11,889 bookings were canceled.
- The cancellation share is large enough to affect revenue, planning, and room allocation.

Core signal: cancellation is a real business problem.

BOOKING STATUS: COUNT + PERCENTAGE

The count and pie chart confirm the scale of the cancellation challenge.



ANALYSIS

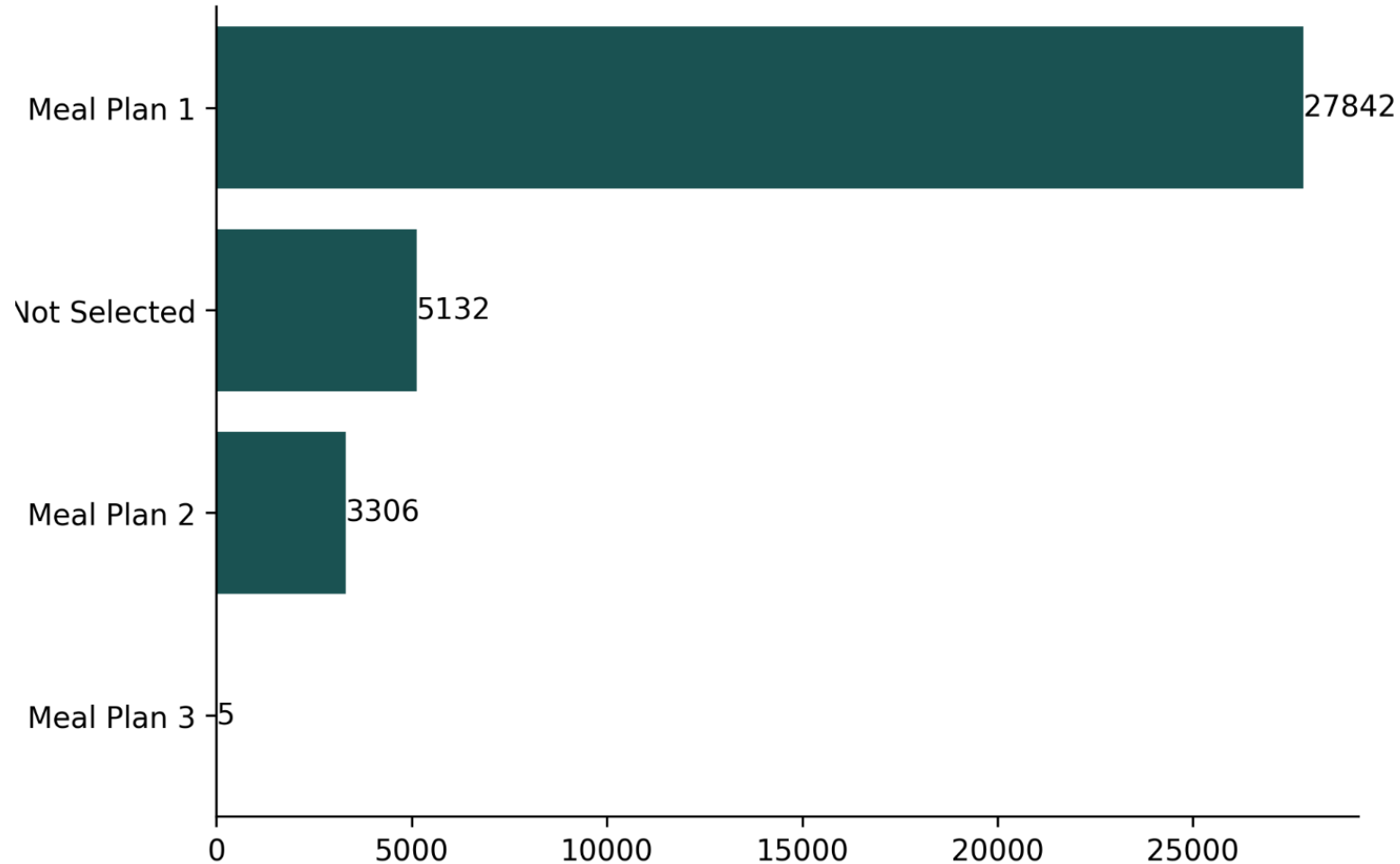
67.23% stayed; 32.77% canceled.

- Most bookings are retained.
- But the canceled group is still big: about one-third of all bookings.
- This makes cancellation prediction valuable for early intervention.

Prediction can help recover risk before arrival.

MEAL PLAN DEMAND

Meal Plan 1 dominates customer meal preferences.



ANALYSIS

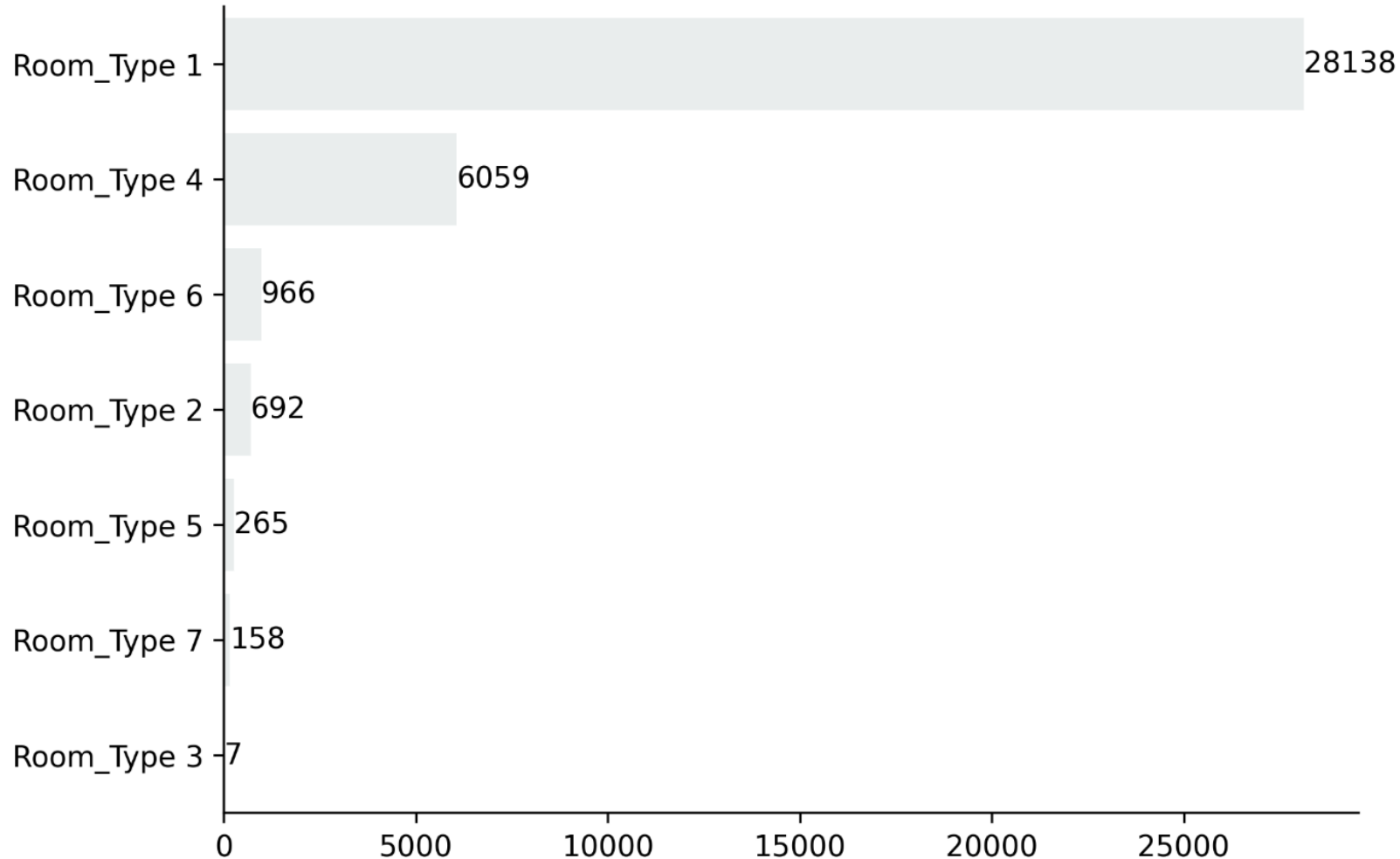
Meal Plan 1 is the clear default.

- Meal Plan 1 records 27,842 bookings.
- “Not Selected” follows with 5,132 bookings.
- Meal Plan 3 is almost unused, with only 5 bookings.

Meal plan may support profiling, but volume is highly concentrated.

ROOM TYPE DEMAND

Room Type 1 is the most demanded room package among the most demanded room packages by Hotel Haven customers.



ANALYSIS

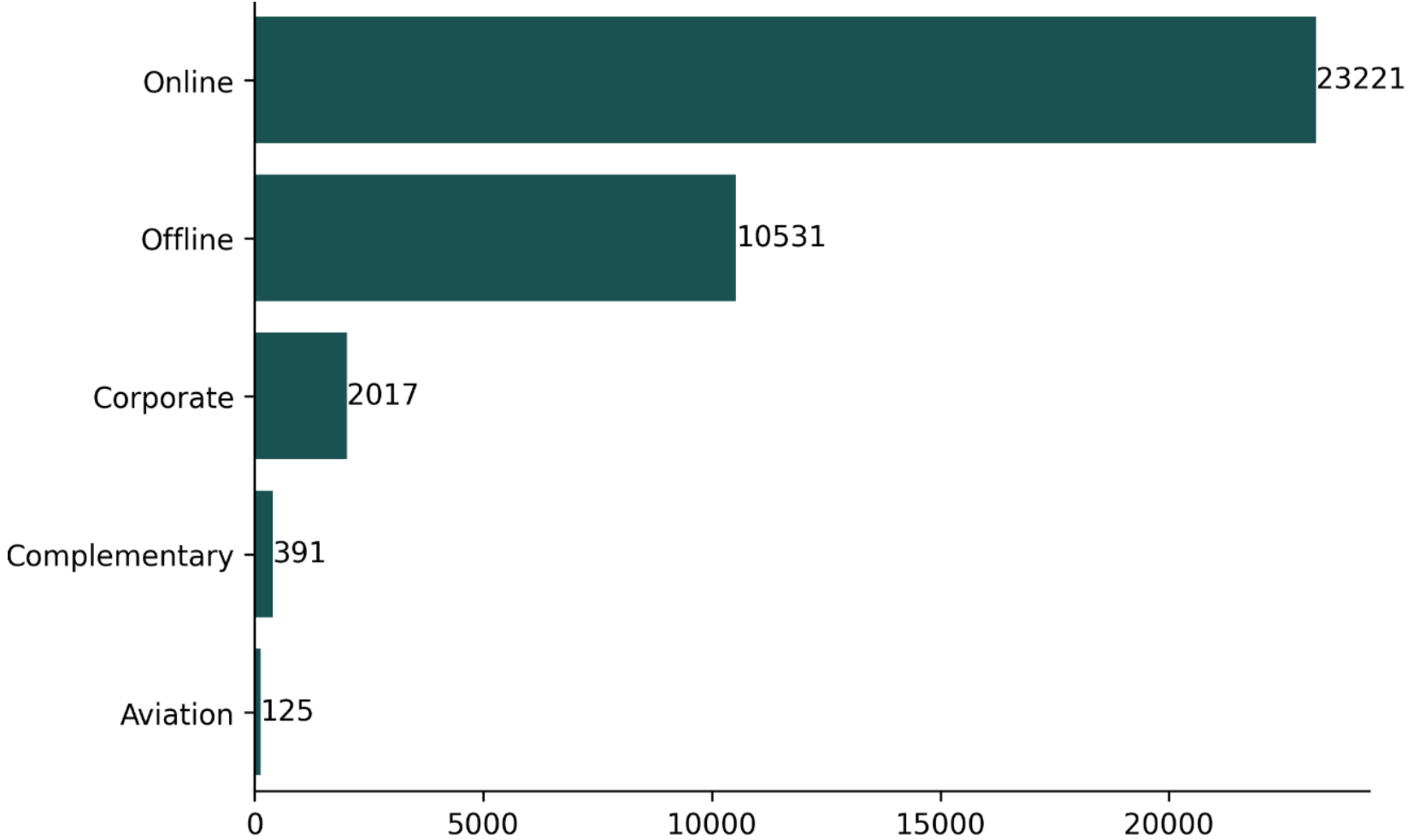
Room Type 1 dominates inventory demand.

- Room Type 1 records 28,138 bookings.
- Room Type 4 is a distant second with 6,059 bookings.
- Other room types have very small volumes compared with Room Type 1.

Demand is concentrated in one main room package.

MARKET SEGMENT DEMAND

Online bookings bring the largest share of Hotel Haven customers.



ANALYSIS

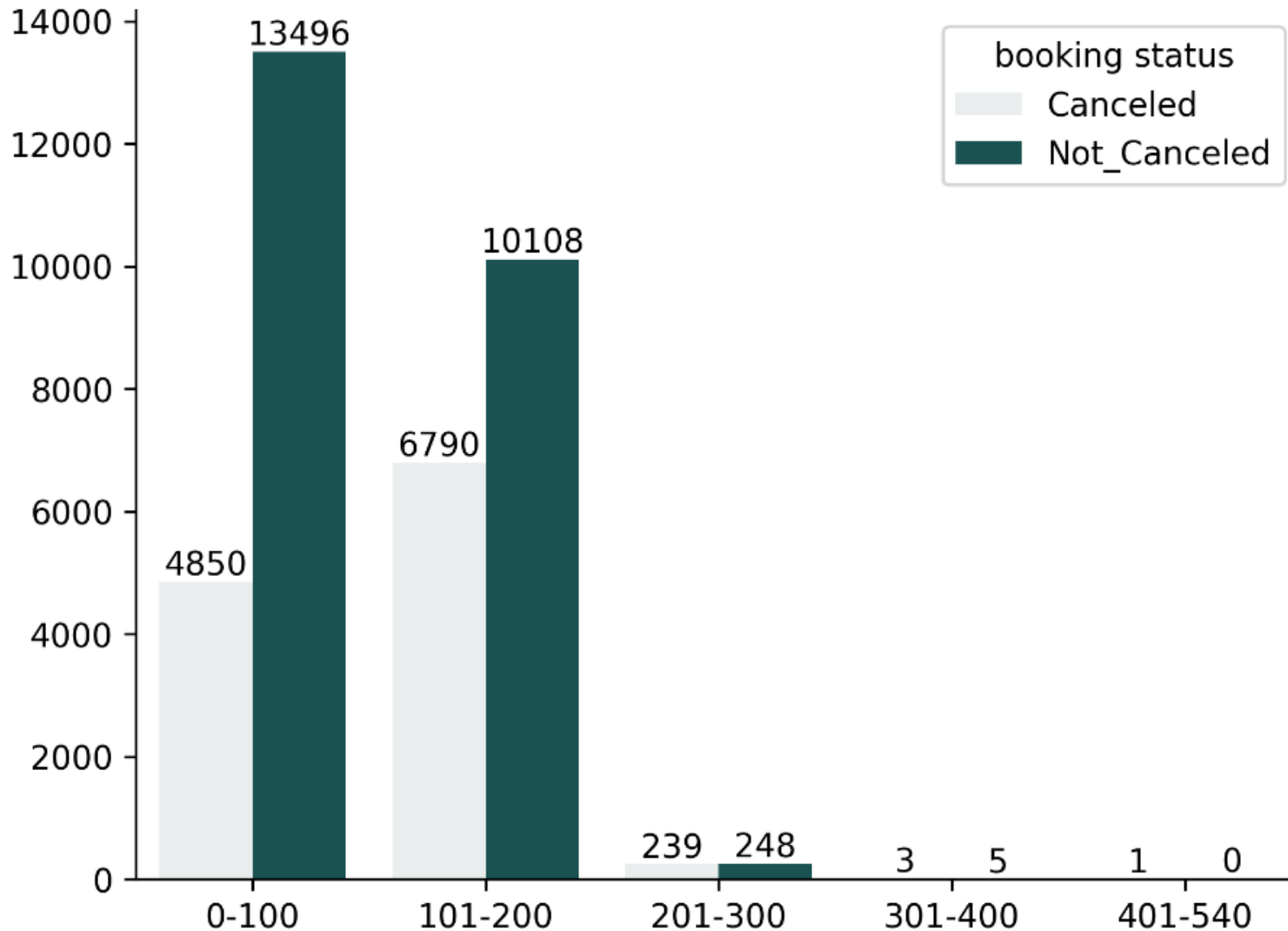
Online is the biggest source of bookings.

- Online bookings account for 23,221 records.
- Offline bookings follow with 10,531 records.
- Corporate, complementary, and aviation segments are much smaller.

Any retention strategy must begin with online bookings.

AVERAGE PRICE RANGE BY BOOKING STATUS

Cancellation behavior is highest around normal and mid-price customers.



ANALYSIS

The 101–200 USD band is the danger zone.

- The 101–200 USD range records the highest cancellation volume.
- The 0–100 USD range also shows high cancellation activity.
- Premium price bands above 300 USD show very low cancellation counts.

Target retention nudges at normal and mid-price guests first.

TOTAL REVENUE BY BOOKING CONDITION

Revenue is stronger among bookings that stayed.

Canceled: 4,297,005.07 (38.0%)
Not_Canceled: 7,054,162.38 (62.0%)



	booking status	total_revenue	percentage
0	Canceled	4297005.07	38.0
1	Not_Canceled	7054162.38	62.0

ANALYSIS

Retained customers carry the revenue story.

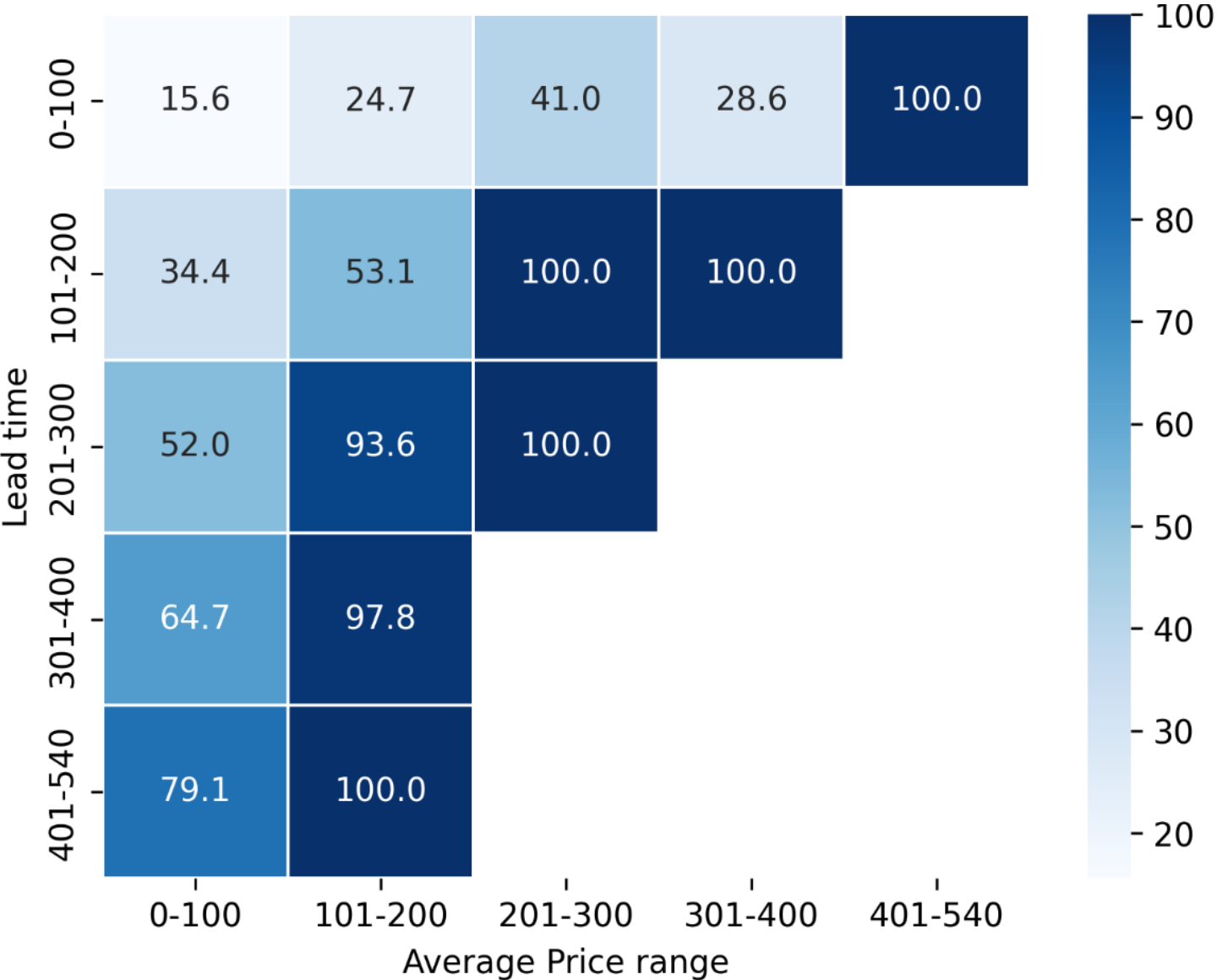
- Not-canceled bookings generated about 62% of total revenue.
- Canceled bookings account for a meaningful 38% revenue exposure.
- Reducing cancellations can protect a major revenue pool.

Saved bookings can directly support revenue protection.

Good news! most the hotel revenue came from loyal customers that about 62% of the revenue is accorded to our loyal customers,

LEAD TIME VS AVERAGE PRICE

The heatmap shows how booking window and price interact.



ANALYSIS

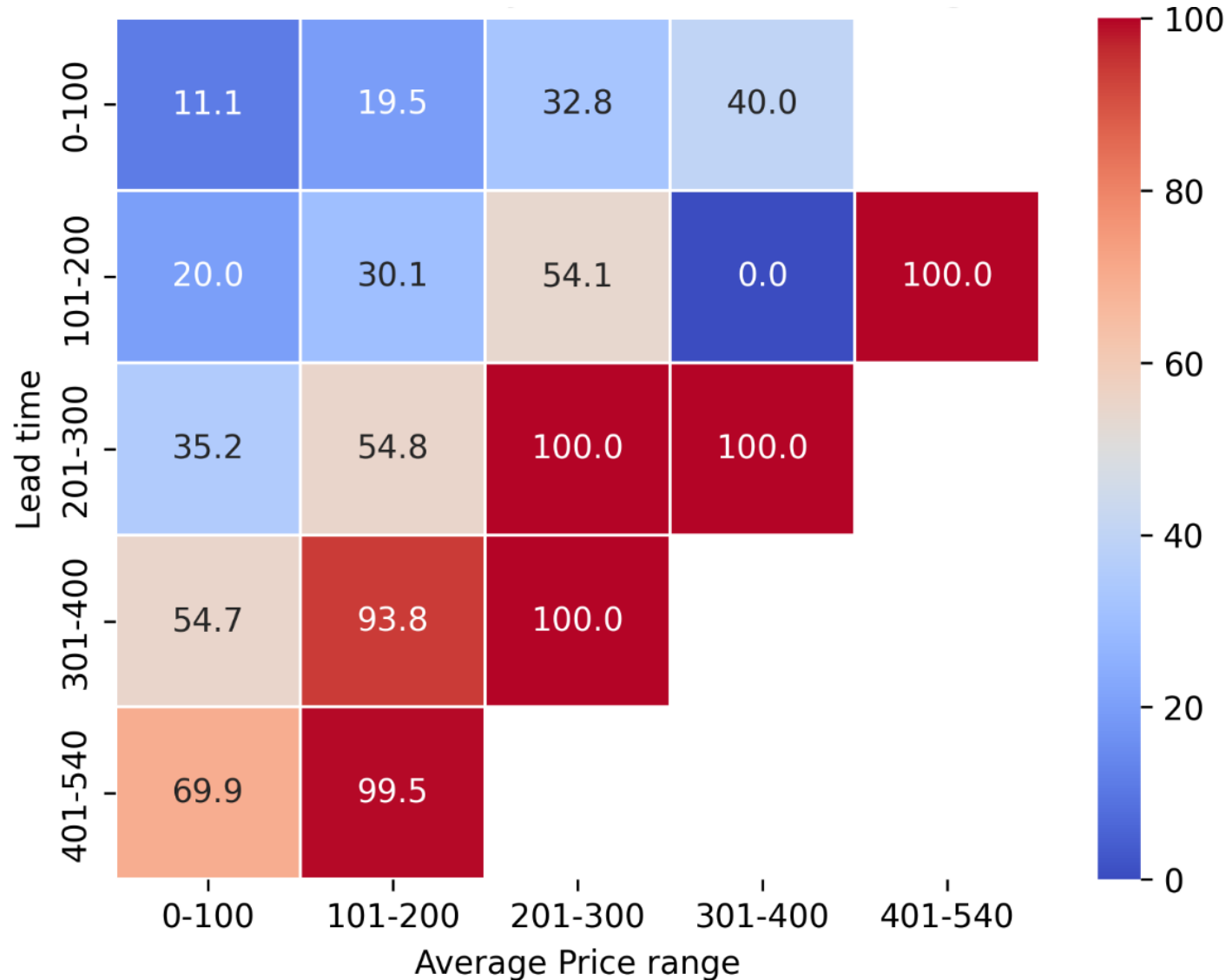
Lead time becomes more dangerous as it rises.

- Cancellation risk intensifies in higher lead-time bands.
- Long booking windows give customers more time to change plans.
- This pattern appears across multiple price ranges.

Lead time is a major cancellation driver.

CANCELLATION RATE BY LEAD TIME + PRICE

The second heatmap confirms the lead-time signal more directly.



ANALYSIS

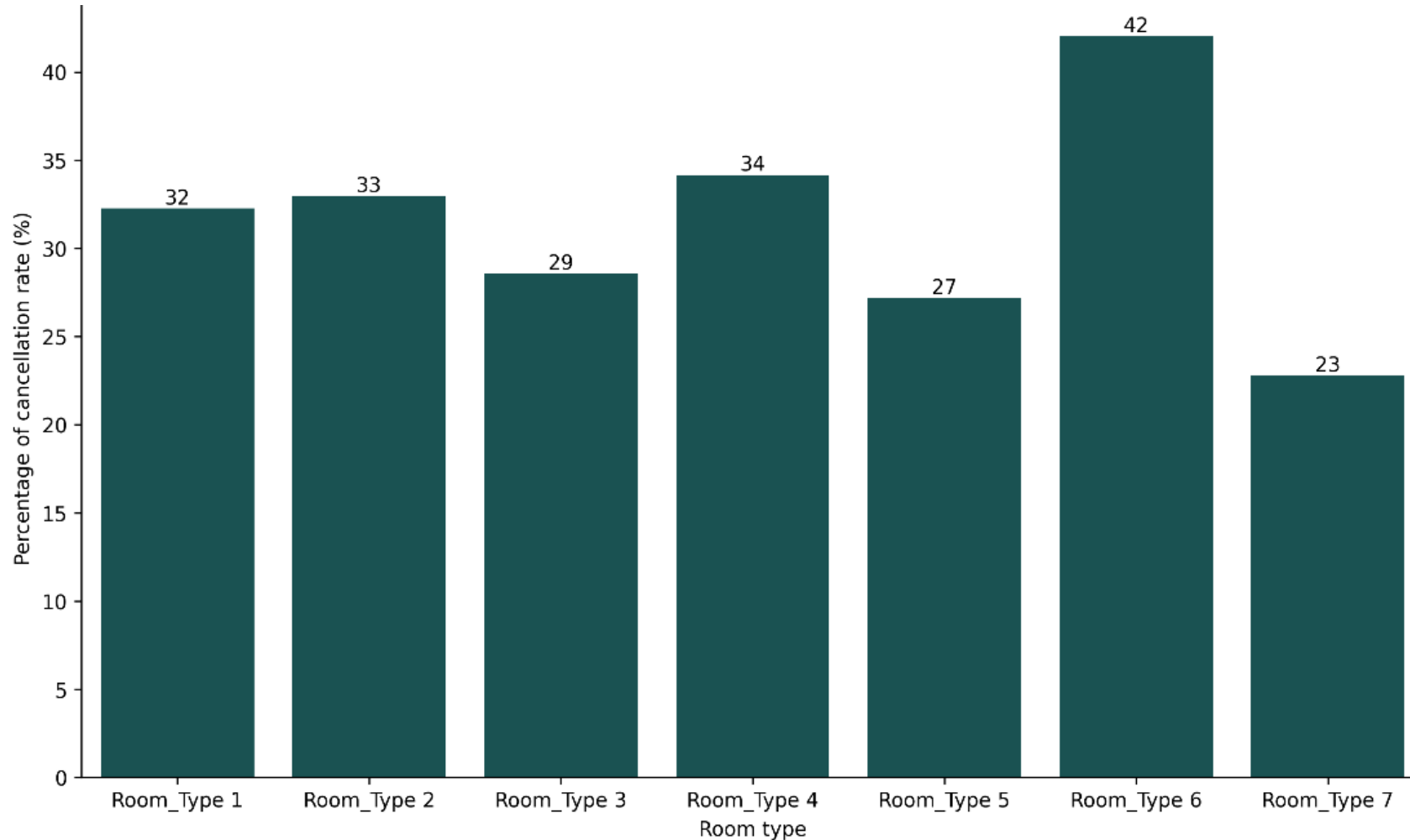
The model should pay close attention to lead time.

- Cancellation rates rise sharply as lead time increases.
- Several higher lead-time and price combinations reach very high risk levels.
- Lead time influences cancellation rate across price categories.

Lead time should be one of the strongest features in the model.

ROOM TYPE AND BOOKING STATUS

Room type has some signal, but the pattern is not fully consistent.



ANALYSIS

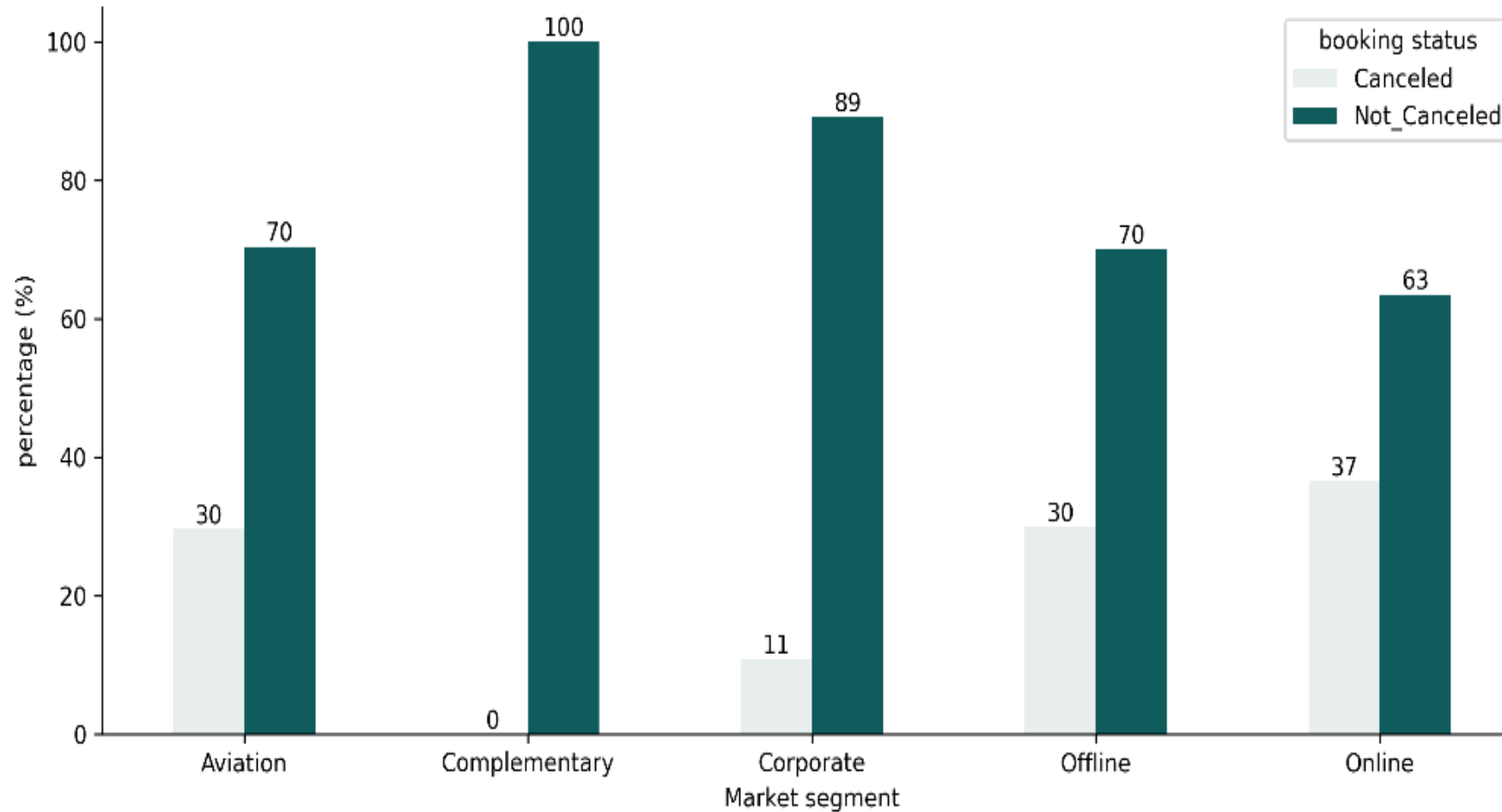
Room type alone is not enough.

- Room Type 6 shows the highest cancellation rate at 42%.
- Room Type 7 is the most reliable, with the lowest rate around 23%.
- Without room descriptions, it is hard to explain why some room types cancel more.

Use room type with other variables, not in isolation.

MARKET SEGMENT AND BOOKING STATUS

Market segment tells us where the risk lives.



ANALYSIS

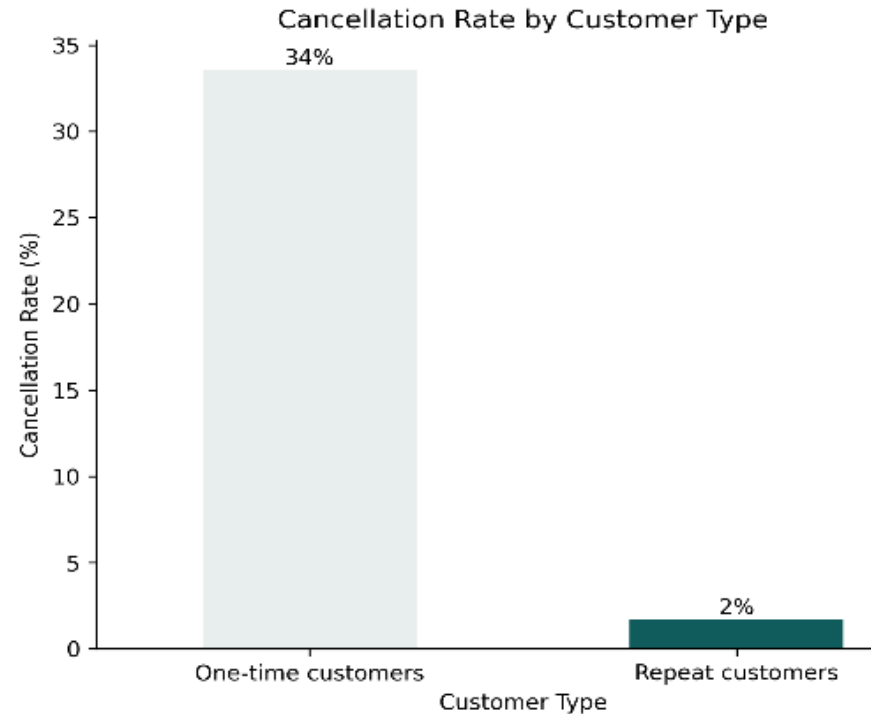
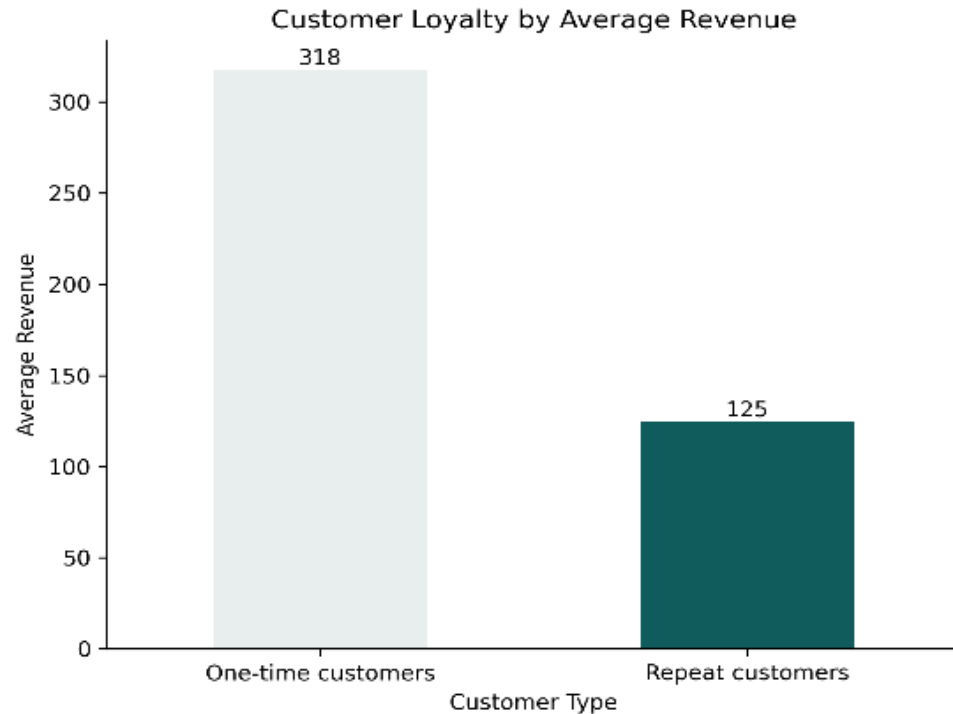
Online is the biggest cancellation-risk segment.

- Online bookings show about 30% cancellation.
- Corporate guests show stronger commitment, with about 11% cancellation.
- Complementary bookings show 0% cancellation in the analysis.

Online bookings need the strongest retention playbook.

CUSTOMER LOYALTY: REVENUE BY CANCELLATION RATE

One-time customers bring revenue but also carry the largest cancellation risk.



ANALYSIS

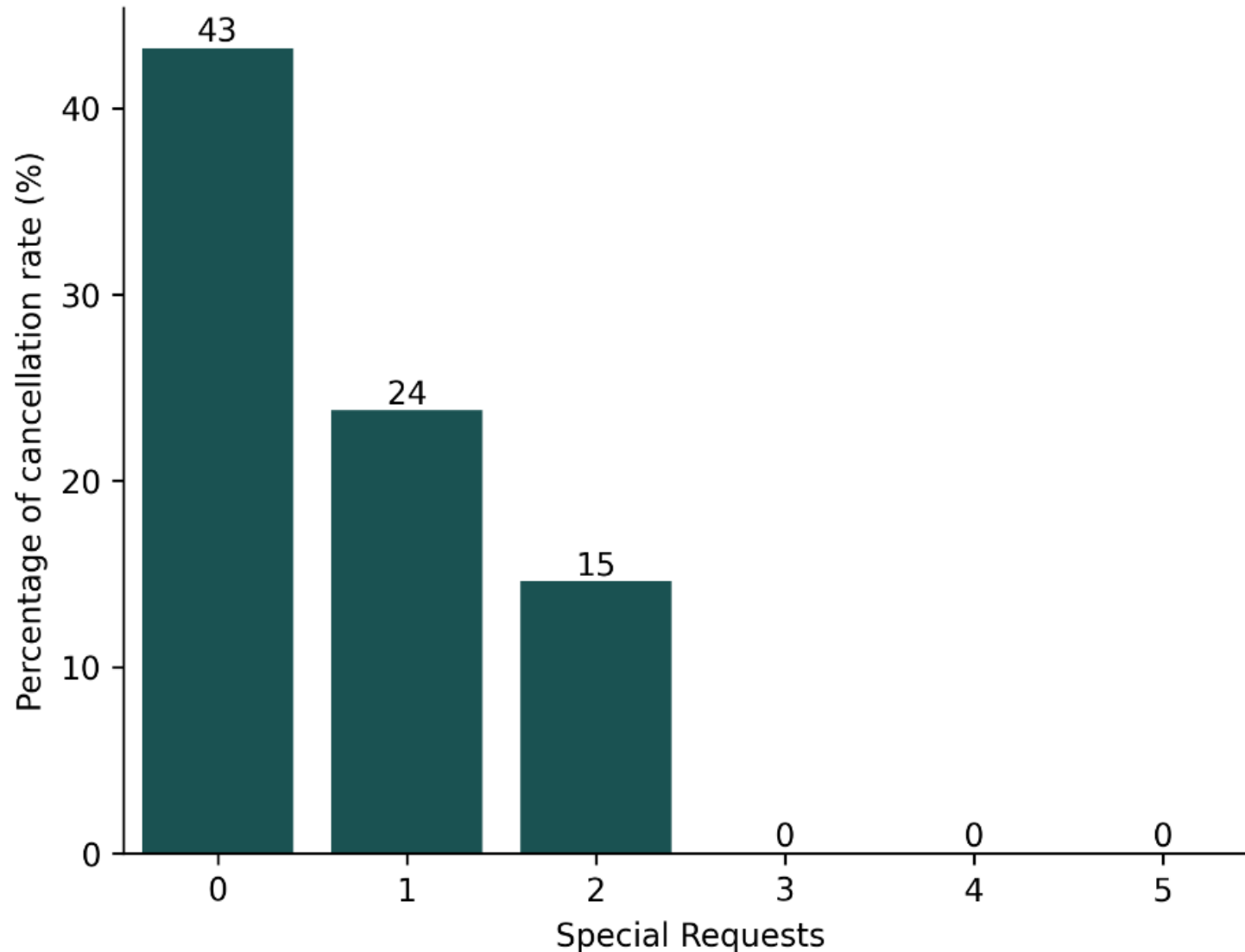
One-time customers need attention.

- One-time customers account for the larger average revenue share.
- They also show a much higher cancellation rate: about 34%.
- Repeat customers are far more stable, with around 2% cancellation.

Customer loyalty is a strong retention signal.

CANCELLATION RATE BY SPECIAL REQUESTS

The more special requests a customer makes, the less likely they are to cancel.



ANALYSIS

Special requests reveal stronger commitment.

- Customers with zero special requests show the highest cancellation rate at 43%.
- One special request reduces cancellation to 24%.
- Two special requests reduce it further to 15%, while 3+ requests record 0% in this sample.

Special requests are a valuable behavioral feature.

MODEL DEPLOYMENT

Model Deployment on Stream lit



Hotel Haven Booking Cancellation Predictor

Predict whether a hotel booking is likely to be cancelled using booking and guest information.



Enter Booking Details

Fill in the booking information below and click the prediction button.

Lead Time

50

Total Guests

2

Average Price Per Room

100

Total Nights

3

Number of Special Requests

1

Repeated Guest?

No

Predict Cancellation



Model Information

Model: Gradient Boosting Model

Task: Booking Cancellation Classification

Output: Canceled or Not Canceled



Business Meaning

This app helps hotels estimate whether a booking may be cancelled, so they can plan room allocation and revenue management more effectively.



Why It Matters

Booking cancellations can affect hotel revenue, staffing, and occupancy planning. Early prediction helps hotels make better operational decisions.

Please enter the booking details and click Predict Cancellation.

From insight to intervention.

The Streamlit app turns booking details into an early cancellation signal the hotel can act on before check-in.

- 1 Lead-time alerts** Trigger early confirmations for long reservation windows.
- 2 Online retention playbook** Send targeted reminders or incentives to online bookings.
- 3 Mid-price protection** Focus on the 101–200 USD segment with high cancellation volume.
- 4 Loyalty conversion** Convert one-time customers into repeat guests with post-booking engagement.

Right booking. Right risk. Right response.



PREDICT CANCELLATIONS EARLY. PROTECT REVENUE FASTER.

Hotel Haven's cancellation model turns booking behavior into a practical early-warning system for retention, planning, and smarter hotel operations.

